

Supplementary Material for Anonymous Submission

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Abstract. This supplementary PDF contains the detailed theoretical analysis omitted from the main ICANN 2026 submission to satisfy the conference page limit. Following the submission guidelines, this file is anonymous and intended to be uploaded as the optional supplementary PDF.

Keywords: Supplementary Material, Stochastic Passivity, Stability Analysis

Scope

ICANN 2026 allows full papers of up to 12 pages including references, while appendices and additional materials may be provided as supplementary files. This document therefore contains the standalone version of the theoretical analysis section on stochastic passivity and stability properties.

1 Theoretical Analysis - Stochastic passivity and stability properties

1.1 Setup and assumptions

We analyze the same discrete-time model used in the main paper:

$$\begin{aligned} x_{t+1} = & x_t + \Delta t (J - R) \nabla H(x_t) + \Delta t u_t \\ & + \sqrt{\Delta t} g_\psi(x_t) \xi_t, \end{aligned} \tag{1}$$

where $u_t := f_{\text{attn}}(x_{t-\tau:t}; \theta)$ denotes the history-dependent attention residual and $\xi_t \sim \mathcal{N}(0, I)$ is independent of the past. Because u_t depends on the full history, the process need not be Markov. Accordingly, all conditional expectations below are taken with respect to the natural filtration $\mathcal{F}_t := \sigma(x_0, \dots, x_t)$, under which u_t is $\mathcal{F}_t$ -measurable.

For later use, define the dissipation and stochastic supply terms

$$\mathcal{D}(x) := \nabla H(x)^\top R \nabla H(x), \tag{2}$$

$$\mathcal{Q}(x) := \frac{1}{2} \text{Tr}(g_\psi(x)^\top \nabla^2 H(x) g_\psi(x)). \tag{3}$$

Assumption 1 (Regularity). The Hamiltonian $H : \mathbb{R}^d \rightarrow \mathbb{R}_+$ belongs to C^2 and has bounded Hessian on the region visited by the trajectory. For the midpoint result below, we additionally assume $H \in C^3$ with bounded third derivatives.

Assumption 2 (Dissipation). The dissipation matrix R is symmetric positive semidefinite and satisfies $R \succeq \alpha I$ for some $\alpha > 0$.

Assumption 3 (Energy coercivity). There exist constants $m_H > 0$ and $b_H \geq 0$ such that

$$\|\nabla H(x)\|^2 \geq m_H H(x) - b_H, \quad x \in \mathbb{R}^d.$$

Assumption 4 (Bounded input and diffusion energy). There exist constants $\bar{u}, \bar{q} > 0$ such that

$$\sup_{t \geq 0} \mathbb{E}[\|u_t\|^2] \leq \bar{u}^2, \quad \sup_{x \in \mathbb{R}^d} \mathcal{Q}(x) \leq \bar{q}.$$

1.2 One-step energy balance and ultimate boundedness

Proposition 1 (Expectation-level one-step energy balance). *Under Assumptions 1 and 4, there exists a constant $C > 0$, independent of Δt , such that the update (1) satisfies*

$$\mathbb{E}[H(x_{t+1}) - H(x_t) \mid \mathcal{F}_t] \leq \Delta t \nabla H(x_t)^\top u_t - \Delta t \mathcal{D}(x_t) + \Delta t \mathcal{Q}(x_t) + C \Delta t^{3/2}. \quad (4)$$

Hence the model is weakly passive in expectation, with dissipation $\mathcal{D}$, deterministic supply $\nabla H(x_t)^\top u_t$, and stochastic supply $\mathcal{Q}$.

Proof. Write the one-step increment as

$$\delta_t := x_{t+1} - x_t = \Delta t ((J - R)\nabla H(x_t) + u_t) + \sqrt{\Delta t} g_\psi(x_t) \xi_t.$$

By the second-order Taylor formula, for some point $\tilde{x}_t$ on the segment between x_t and x_{t+1} ,

$$H(x_{t+1}) - H(x_t) = \nabla H(x_t)^\top \delta_t + \frac{1}{2} \delta_t^\top \nabla^2 H(\tilde{x}_t) \delta_t.$$

Taking conditional expectation with respect to $\mathcal{F}_t$ and using $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$ and $\mathbb{E}[\xi_t \xi_t^\top \mid \mathcal{F}_t] = I$, the first-order term becomes

$$\begin{aligned} \mathbb{E}[\nabla H(x_t)^\top \delta_t \mid \mathcal{F}_t] &= \Delta t \nabla H(x_t)^\top (J - R)\nabla H(x_t) + \Delta t \nabla H(x_t)^\top u_t \\ &= -\Delta t \mathcal{D}(x_t) + \Delta t \nabla H(x_t)^\top u_t, \end{aligned}$$

because $a^\top J a = 0$ for any skew-symmetric J .

For the second-order term, expand $\delta_t \delta_t^\top$. The dominant contribution comes from the diffusion covariance,

$$\mathbb{E}[\delta_t \delta_t^\top \mid \mathcal{F}_t] = \Delta t g_\psi(x_t) g_\psi(x_t)^\top + \mathcal{O}(\Delta t^{3/2}),$$

while all drift-drift and drift-diffusion cross terms are of order at least $\Delta t^{3/2}$. Since $\nabla^2 H$ is bounded,

$$\frac{1}{2} \mathbb{E}[\delta_t^\top \nabla^2 H(\tilde{x}_t) \delta_t \mid \mathcal{F}_t] \leq \Delta t \mathcal{Q}(x_t) + C \Delta t^{3/2}.$$

Combining the first- and second-order terms yields (4).

Theorem 5 (Ultimate boundedness in expectation). *Under Assumptions 1–4, there exist constants $\beta > 0$ and $c_0 > 0$, independent of t , such that for all sufficiently small Δt ,*

$$\mathbb{E}[H(x_{t+1})] \leq (1 - \beta \Delta t) \mathbb{E}[H(x_t)] + c_0 \Delta t. \quad (5)$$

Consequently,

$$\limsup_{t \rightarrow \infty} \mathbb{E}[H(x_t)] \leq \frac{c_0}{\beta} = \mathcal{O}\left(\frac{\bar{u}^2 + \bar{q} + b_H + 1}{\alpha m_H}\right). \quad (6)$$

Proof. Taking expectation in (4) gives

$$\mathbb{E}[H(x_{t+1})] \leq \mathbb{E}[H(x_t)] + \Delta t \mathbb{E}[\nabla H(x_t)^\top u_t] - \Delta t \mathbb{E}[\mathcal{D}(x_t)] + \Delta t \bar{q} + C \Delta t^{3/2}.$$

Using Young's inequality with parameter $\alpha/2$,

$$\nabla H(x_t)^\top u_t \leq \frac{\alpha}{2} \|\nabla H(x_t)\|^2 + \frac{1}{2\alpha} \|u_t\|^2,$$

and since $R \succeq \alpha I$,

$$\mathcal{D}(x_t) = \nabla H(x_t)^\top R \nabla H(x_t) \geq \alpha \|\nabla H(x_t)\|^2.$$

Therefore

$$\nabla H(x_t)^\top u_t - \mathcal{D}(x_t) \leq -\frac{\alpha}{2} \|\nabla H(x_t)\|^2 + \frac{1}{2\alpha} \|u_t\|^2.$$

Applying Assumption 3,

$$-\frac{\alpha}{2} \|\nabla H(x_t)\|^2 \leq -\frac{\alpha m_H}{2} H(x_t) + \frac{\alpha b_H}{2},$$

and using $\mathbb{E}[\|u_t\|^2] \leq \bar{u}^2$, we obtain

$$\mathbb{E}[H(x_{t+1})] \leq \left(1 - \frac{\alpha m_H}{2} \Delta t\right) \mathbb{E}[H(x_t)] + \Delta t \left(\frac{\bar{u}^2}{2\alpha} + \bar{q} + \frac{\alpha b_H}{2}\right) + C \Delta t^{3/2}.$$

For sufficiently small Δt , the last term can be absorbed into the Δt -term, giving (5) with $\beta = \alpha m_H/2$. Iterating the linear recursion and taking $t \rightarrow \infty$ yields (6).

Remark 1. Theorem 5 uses the full filtration $\mathcal{F}_t$, so the history dependence of the attention term does not invalidate the argument. What matters is not Markovianity, but the adaptedness and second-moment control of u_t .

1.3 Implicit midpoint discretization

We next consider the continuous-time stochastic port-Hamiltonian model

$$dx = ((J - R)\nabla H(x) + u(t)) dt + g_\psi(x) \circ dW_t, \quad (7)$$

interpreted in the *Stratonovich* sense. This interpretation is natural for energy calculations because it obeys the ordinary chain rule $dH(x_t) = \nabla H(x_t)^\top \circ dx_t$, and the implicit midpoint rule is a standard structure-preserving discretization for Stratonovich systems.

Let u_t denote the discrete input over one time step, and define the midpoint state $\bar{x}_t := \frac{1}{2}(x_t + x_{t+1})$. The implicit midpoint scheme reads

$$\begin{aligned} x_{t+1} = x_t + \Delta t ((J - R)\nabla H(\bar{x}_t) + u_t) \\ + \sqrt{\Delta t} g_\psi(\bar{x}_t) \xi_t. \end{aligned} \quad (8)$$

Proposition 2 (Weak passivity of the implicit midpoint discretization).

Assume $H \in C^3$ with bounded third derivatives and that u_t and g_ψ have bounded second moments along the trajectory. Then there exists $C > 0$ such that the midpoint update (8) satisfies

$$\begin{aligned} \mathbb{E}[H(x_{t+1}) - H(x_t)] \leq \Delta t \mathbb{E}[\nabla H(\bar{x}_t)^\top u_t] \\ - \Delta t \mathbb{E}[\mathcal{D}(\bar{x}_t)] + \Delta t \mathbb{E}[\mathcal{Q}(\bar{x}_t)] + C \Delta t^2. \end{aligned} \quad (9)$$

Proof. Let $\delta_t := x_{t+1} - x_t$. By the midpoint expansion for C^3 functions,

$$H(x_{t+1}) - H(x_t) = \nabla H(\bar{x}_t)^\top \delta_t + R_t, \quad |R_t| \leq C \|\delta_t\|^3.$$

Substituting (8) gives

$$\begin{aligned} \nabla H(\bar{x}_t)^\top \delta_t &= \Delta t \nabla H(\bar{x}_t)^\top (J - R) \nabla H(\bar{x}_t) \\ &\quad + \Delta t \nabla H(\bar{x}_t)^\top u_t \\ &\quad + \sqrt{\Delta t} \nabla H(\bar{x}_t)^\top g_\psi(\bar{x}_t) \xi_t. \end{aligned}$$

Again, the conservative part vanishes because $a^\top J a = 0$, so the deterministic contribution equals

$$-\Delta t \mathcal{D}(\bar{x}_t) + \Delta t \nabla H(\bar{x}_t)^\top u_t.$$

For the stochastic work term, the Stratonovich midpoint rule has the standard weak expansion

$$\mathbb{E}[\sqrt{\Delta t} \nabla H(\bar{x}_t)^\top g_\psi(\bar{x}_t) \xi_t] = \Delta t \mathbb{E}[\mathcal{Q}(\bar{x}_t)] + \mathcal{O}(\Delta t^2),$$

which is precisely the chain-rule correction associated with the diffusion. Moreover, $\delta_t = \mathcal{O}(\Delta t^{1/2})$ in L^p , so $\mathbb{E}|R_t| \leq C \Delta t^{3/2}$. In the weak midpoint expansion, this remainder is dominated by the $\mathcal{O}(\Delta t^2)$ local defect. Collecting all terms yields (9).

Remark 2. We intentionally state the midpoint defect as $\mathcal{O}(\Delta t^2)$ rather than $\mathcal{O}(\Delta t^3)$. Under the assumptions used here, the $\mathcal{O}(\Delta t^2)$ weak defect is the relevant robust statement and is sufficient for the qualitative passivity interpretation.